

2.2 Vanishing and Exploding Gradients

Backprop through time is a product of Jacobians. Long-range signal dies or blows up.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

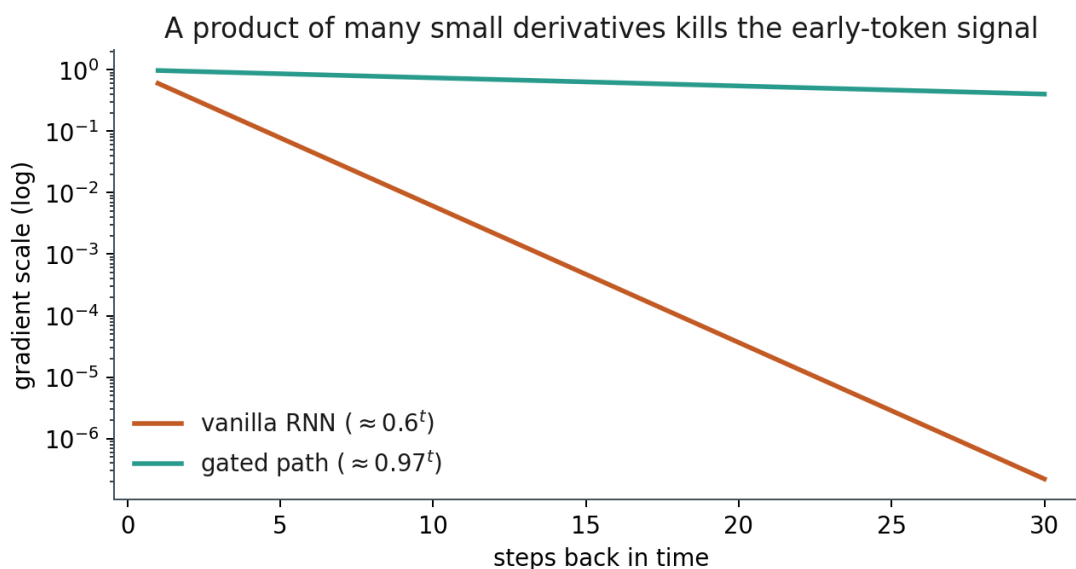
Backprop through an unrolled RNN multiplies one Jacobian per time step. If those matrices shrink, the early-token gradient **vanishes**. If they grow, it **explodes**. Either way, long-range learning dies.

1. A product along the chain

The loss at the last token, $L(h_T)$, depends on h_1 through

$$\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial h_T} \frac{\partial h_T}{\partial h_{T-1}} \cdots \frac{\partial h_2}{\partial h_1}.$$

Each factor $\partial h_t / \partial h_{t-1}$ contains W_h and $\tanh'$. For $\tanh$, $|\tanh'| \leq 1$, and it is $\ll 1$ unless the pre-activation sits near 0. A product of twenty numbers smaller than 1 is tiny.



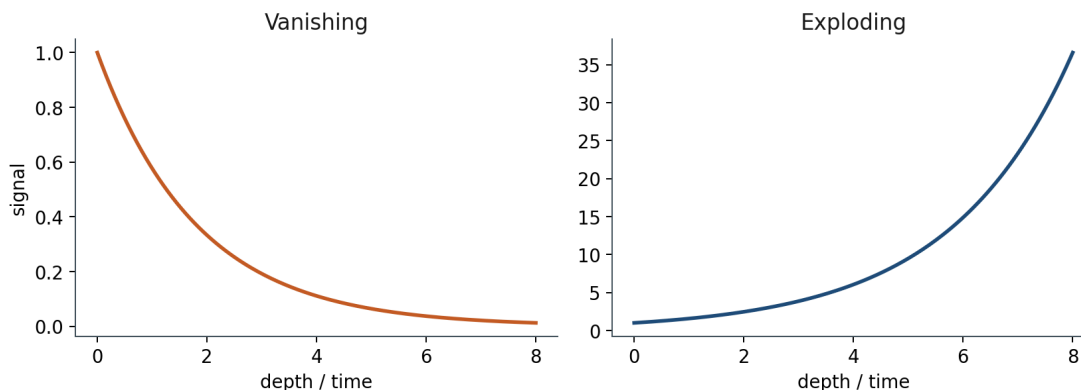
The gated curve is the slogan for LSTM/GRU: keep a path whose multiplier can stay near 1.

In 1-D, if each factor is 0.5, ten steps give $0.5^{10} \approx 0.001$. The early token still sits in the forward state as a faint trace, but the **learning signal** is gone.

2. Vanishing versus exploding

Same product, opposite problem. If $\|W_h\|$ is large, the signal blows up, weights become NaNs, and training stops. **Gradient clipping** (rescale if the global norm exceeds a threshold) is the usual bandage

for exploding. It does not fix vanishing.



Clipping: if $\|g\| > c$, replace g by $c g / \|g\|$. You shrink a huge vector. A vanished vector is already near 0; multiplying it by 1 does nothing useful.

3. What this means for language

A vanilla RNN can learn “the next word looks like the last few.” It struggles with “the negation was at the start of the review.” Transformers side-step the chain: every token attends to every other token in one step (Week 3). Gates (Week 2.3) were the fix *inside* the recurrent family.

In a project, if you still train an RNN on long documents, watch the gradient norm. If it is always ≈ 0 on early positions, the net is not using the beginning of the sequence.

Lab 2 is a copy task: the first bit must survive many steps. Vanilla RNNs fail as the gap grows; that is this product, not a PyTorch bug.

4. Teaching this note

30–40 minutes. Write the product of Jacobians, then the 1-D geometric example (0.5^T vs 1.5^T). Draw clipping as a cap on a tall arrow, not a resurrection of a dead one. Play the vanishing/exploding segment of CS224N L6 (about **45:00–58:00**; if chapter marks differ, jump to the Jacobian-product board). LSTM details wait for 2.3.

5. Worked example

Scalar RNN: $h_t = \tanh(wh_{t-1})$. Then $h'_t(h_{t-1}) = \tanh'(z_t) w$ with $|\tanh'| \leq 1$.

Take $w = 0.5$ and pretend $\tanh' = 1$ (best case). The factor per step is 0.5:

$$0.5^5 = 0.03125, \quad 0.5^{10} \approx 0.0010, \quad 0.5^{20} \approx 9.5 \times 10^{-7}.$$

Now $w = 2$: $2^{10} = 1024$. Same chain, opposite disaster.

Clipping: gradient $g = 50$, threshold $c = 5$. New gradient 5. If instead $g = 10^{-6}$, clipping leaves it at 10^{-6} .

6. Where students get stuck

- Thinking clipping “fixes RNNs,” including vanishing.
 - Forgetting $\tanh' \leq 1$ and blaming only W_h .
 - Plotting loss and missing a **per-position** gradient plot in Lab 2.
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7. Video

Watch [Stanford CS224N Lecture 6 \(RNNs / vanishing_gradients\)](#).

Play the vanishing and exploding segment (roughly **45:00–58:00**). Pause on the product of Jacobians and on the plot of gradient size vs. steps back in time. Skip the rest of the lecture hour unless you want n-gram language modeling as optional homework.

8. Practice

1. Why is $\tanh'$ a problem when the state saturates at ± 1 ?
2. Clipping stops explosions. Why does it not restore a vanished signal?
3. In one sentence: what would you plot in TensorBoard to see this failure in Lab 2?
4. If each Jacobian factor has operator norm 0.8, the bound on $\|\partial h_{20} / \partial h_1\|$ is 0.8^{19} . Compute 0.8^{10} and 0.8^{19} (calculator is fine). Is the 19-step bound closer to 0.1 or to 0.01?
5. A clip threshold $c = 1$ meets a gradient of $[-6, 8]$ (Euclidean norm 10). Write the clipped vector.